SubSLD: Evidence-Paired Generation of Substation Single-Line Diagrams from Volunteered Geographic Information

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Abstract*—Public models of national transmission networks stop at the substation boundary: what lies inside—busbars, bays, transformers, circuit counts—is proprietary and therefore absent from open datasets. We present *SubSLD*, a method that reconstructs the internal composition of substations from volunteered geographic information alone and renders it as an *evidence-paired figure*: site geometry over aerial imagery beside a machine-drawn single-line diagram. The method is formulated as an evidence-closure operator whose output contains only elements possessing an observational witness, so that soundness ("no fabrication") holds by construction; terminal binding is a lexicographic maximisation over an ordered evidence vocabulary; circuit counts are produced by a certified lower-bound estimator; and flow direction is a three-valued inference that abstains explicitly rather than guessing. Applied to all of Japan, the pipeline extracts 7,239 substation structures with 47,979 evidence-bound terminals in about four seconds and renders paired figures for every site. We report coverage rather than claim completeness: circuit evidence reaches $68.2\,\%$ of 40,087 lines, busbar ways are mapped at only $14.2\,\%$ of sites (from $53.3\,\%$ in Hokkaido to $5.1\,\%$ in Okinawa), and direction inference abstains on $39.4\,\%$ of line groups. A photointerpretation review of fourteen transmission substations lacking busbar ways finds that $64\,\%$ are mappable omissions rather than physical absences, turning the measured gap into an actionable contribution list for the source database. The extracted structure is exported to CIM/CGMES—4,743 `BusbarSections` and 8,475 `Bays` nationwide, with inferred elements flagged in their `description`—so the result is consumable by standard power-system tooling. All artefacts are regenerated by a single command and are published online.

***Index Terms*—Power system modelling, substations, single-line diagram, OpenStreetMap, volunteered geographic information, open data, provenance.**

I. Introduction

OPEN models of electric power networks have advanced quickly: continent-scale datasets derived from OpenStreetMap (OSM) now support transmission-level studies, and several national models are published with electrical parameters attached. These models share a boundary, however. They describe how substations are *connected*, but not what a substation *is*: the busbar sections it contains, how incoming lines attach to them, how many circuits each line carries, and which voltage levels are joined by transformers. That information sits in operator drawings and is not public.

The gap matters for two reasons. First, several analyses—bus-splitting, breaker-level contingency screening, and hosting-capacity work at the bay level—require internal structure that connectivity models cannot supply. Second, and more subtly, the boundary hides the *quality* of the open data: when a model is drawn only as nodes and edges, one cannot see whether a substation's representation rests on strong evidence or on a loose spatial heuristic.

This paper takes the position that the interior of substations *is* partly observable in open data, provided the observation is treated with discipline. Aerial imagery shows outdoor busbars and bay rows; OSM contains substation polygons, in-yard ways tagged `line=busbar` or `line=bay`, and transmission line ways carrying `circuits`, `cables` and `wires` tags. What has been missing is not data but a method that (i) asserts only what evidence supports, (ii) makes the strength of that evidence visible in the output, and (iii) reports its own coverage as a result rather than hiding it.

We propose **SubSLD** (evidence-paired substation single-line diagramming). Its contributions are:

- **C1.** A formulation of substation structure extraction as an *evidence-closure operator* (Section IV), with determinism, idempotence and soundness established by construction; the complement—completeness—is deliberately not claimed but measured.
- **C2.** A *certified lower-bound estimator* for circuit and conductor counts (Section V), which formalises the rule "never fill an untagged value by guessing" as a guarantee that aggregates never overstate the physical asset.
- **C3.** A *three-valued direction inference* with explicit abstention (Section VI), so that unresolved cases are rendered as unresolved instead of being silently resolved.
- **C4.** A nationwide instantiation—7,239 substations, paired figures for every site, an in-browser viewer—together with a coverage evaluation and a photointerpretation study that converts the measured gaps into an OSM contribution list (Sections VII–IX).
- **C5.** A *standards-conformant serialisation* (Section VIII): the node-breaker structure maps one-to-one onto CIM/CGMES classes, and the provenance of inferred elements survives the export, so downstream tools receive both the model and its caveats.

II. Related Work

OSM-derived power network models are well established at transmission level. Continental extractions such as SciGRID and PyPSA-Eur build node–edge graphs and attach electrical parameters by voltage-class heuristics; national efforts follow

TABLE I
NOTATION

Symbol	Meaning
O	observation (polygons, ways, tags)
S^*	extracted structure (node-breaker)
$\mathrm{Poly}(s)$	site footprint of substation s
$t \in \mathrm{Terminals}(s)$	line end bound to s, carrying its evidence
$kv_v,\ L(s,v)$	voltage level magnitude and its bound lines
$\hat{c}(w)$	circuit-count estimate for way w
$\mathrm{far}(g)$	remote substations of line group g

the same pattern. All of them abstract a substation to a single node. Where internal structure appears in open modelling, it is usually *assumed* (a single collector bus per voltage level) rather than extracted, because no observational source was thought to exist.

Two adjacent literatures inform our approach. In data quality research for volunteered geographic information, intrinsic indicators—completeness by comparison against reference data, tag coverage, contributor density—are used to characterise a dataset without ground truth; we adopt this stance and report coverage as a primary result. In remote sensing, detection of transmission infrastructure from aerial imagery (towers and conductors) is an active topic, with public datasets such as TTPLA; we use imagery in a lighter role, as a human verification and triage channel rather than as an automatic detector, and we mark it as a distinct evidence class when it is used.

Our formulation is closest in spirit to provenance-carrying data integration: each asserted element retains the witness that justified it, and the rendering layer is required to display evidence strength. To our knowledge, no prior work produces machine-generated single-line diagrams for a national fleet of substations from open data with per-element provenance.

III. Problem Setting

The input is a regional OSM extract: substation features (polygons or points), line ways with geometry and tags, and in-yard ways. Coordinates are quantised to approximately one metre,

$$q(p) = \big(\mathrm{round}(\phi, 5),\ \mathrm{round}(\lambda, 5)\big), \tag{1}$$

which fuses duplicate registrations of the same facility, and the per-island coordinate graph $G_I = (V_I, E_I)$ is formed over the nodes whose region belongs to synchronous island I.

For a site s we write $\mathrm{Poly}(s)$ for its footprint, W_s for the in-yard ways it contains, W_ℓ for the geometry of line ℓ, and $\mathrm{Terminals}(s)$ for the set of line ends bound to s. A voltage level is denoted v with nominal magnitude kv_v, and $L(s, v)$ is the set of lines bound to that level. Table I summarises the notation.

IV. Formulation: Structure Extraction as Evidence Closure

Definition 1 (Evidence closure): Let $\mathrm{witnesses}(x, O)$ be the set of observations in O that support the existence of structural element x. The extraction operator is

$$F(O) \;=\; S^* \;=\; \big\{\, x \;\big|\; \mathrm{witnesses}(x, O) \neq \varnothing \,\big\}, \tag{2}$$

i.e. the largest structure all of whose elements are witnessed.

This is the method proper; the algorithm below is one way to compute it.

Proposition 1 (Properties of F): (i) Determinism: F is a function—identical input yields byte-identical structure (verified nationally by a regeneration test). *(ii) Idempotence:* re-application does not change the structure, so the extraction may be embedded in a repeatedly executed regeneration pipeline. *(iii) Soundness:* every element of S^* possesses a witness. This is immediate from the construction and is the formal content of the "no fabrication" rule.

Completeness—that every physically existing element appears in S^*—does *not* hold, because it is bounded by the coverage of O. We therefore do not claim it; we measure it (Section VII).

A. Terminal binding as lexicographic maximisation

Line ends are attached to structure through an ordered evidence vocabulary,

$$\text{vertex} \;\succ\; \text{polygon} \;\succ\; \text{lead-in}, \tag{3}$$

with the predicates

$$\text{vertex}(t):\ \exists\, v \in W_\ell \cap W_s, \tag{4}$$

$$\text{polygon}(t):\ p_{\text{end}}(\ell) \in \mathrm{Poly}(s), \tag{5}$$

$$\text{lead-in}(t):\ d\big(p_{\text{end}}(\ell), \mathrm{Poly}(s)\big) \leq \delta_{\text{lead}}, \tag{6}$$

where $\delta_{\text{lead}} = 0.6\,\text{km}$. The binding chosen for t is

$$\text{binding}(t) = \max\nolimits_{\succ} \{\, e \mid e \text{ witnesses } t \,\}, \tag{7}$$

and it is retained in the record, so that weak evidence can be drawn weakly (lead-in bindings are rendered as dashed stubs).

B. Consistency constraint

Physical proximity is not electrical connection. Shared corridors and parallel routes place lines of different classes within metres of one another, so the closure is restricted by a voltage-consistency gate,

$$\text{admit}(\ell, n) \iff \text{witnesses} \neq \varnothing \wedge \neg\big(|kv_\ell - kv_n| > 0.25 \max(kv_n, 1)\big). \tag{8}$$

In our campaign this gate correctly rejected a $66\,\text{kV}$ fragment touching a $154\,\text{kV}$ junction within $80\,\text{m}$; subsequent diagnosis showed the fragment itself to be a cross-region registration artefact, i.e. the gate prevented an error whose true cause lay elsewhere in the data.

C. Algorithm

Algorithm 1 realises F. The final step deserves comment: in gas-insulated and indoor substations a busbar is not a visible way and can never be observed, yet its existence is electrically necessary when two or more lines bind strongly to the same level. We therefore emit an *inferred* busbar, recorded with `source = inferred-topology` and rendered dashed. This asserts existence only—no geometry, no terminal re-binding—so soundness in the sense of Proposition 1 is preserved with the caveat made visible.

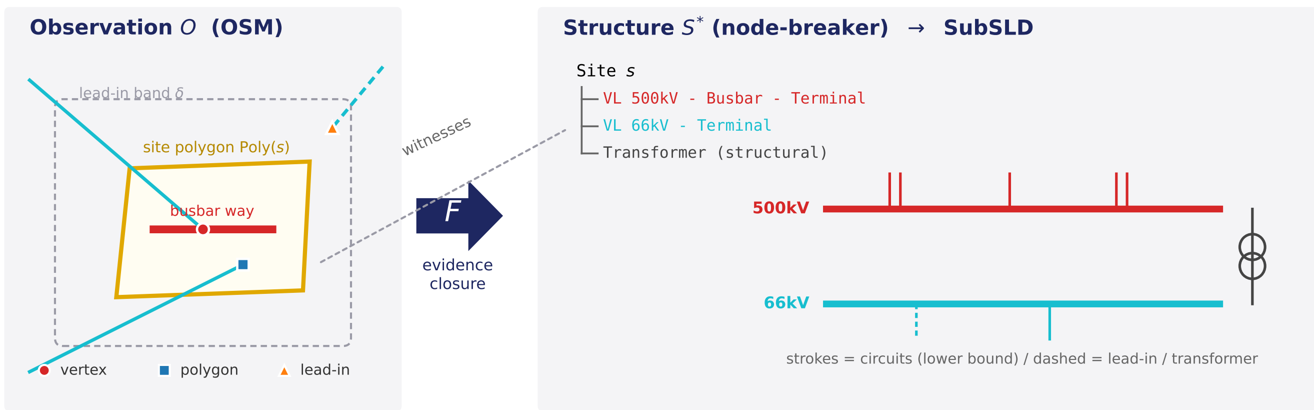


Fig. 1. The SubSLD method. Observation O (site polygon, in-yard busbar way, line ends with their binding evidence, lead-in band δ) is mapped by the evidence closure F to a node-breaker structure S^*, which is rendered as the paired figure. Every drawn element retains the witness that justified it.

Algorithm 1 Structure extraction for one substation

Require: site feature s, preprocessed ways W

Poly$(s) \leftarrow$ shape(s); $V \leftarrow$ vcls(tags) $\cup$ vcls(in-yard)
for all vertex-connected components of $W[\texttt{busbar}]$ **do**
  emit BusbarSection (derive kv from adjacency if untagged)
end for
for all vertex-connected components of $W[\texttt{bay}]$ **do**
  emit Bay, recording the busbars it touches
end for
for all end points p of line ways **do**
  $b \leftarrow \max_{\succ}\{\text{vertex}, \text{polygon}, \text{lead-in}\}$
  if b exists and admit holds **then**
    emit Terminal$(v,\ \text{attach},\ b,\ \text{par} = \hat{c}(w))$
  end if
end for
emit Transformer for adjacent voltage-class pairs (structural)
if a level has ≥ 2 strongly-bound terminals and no busbar way **then** emit an *inferred* busbar (marked)

Ensure: S^* with a witness on every element

V. Estimation of Circuit and Conductor Counts

OSM line ways carry `circuits`, `cables` and `wires` tags irregularly. Rather than imputing missing values, we define

$$\hat{c}(w) = \begin{cases} c_{\text{tag}}(w), & \text{if } \texttt{circuits} \text{ is present,} \\ \lfloor n_{\text{cables}}(w)/3 \rfloor, & \text{else if } \texttt{cables} \text{ is present,} \\ 1, & \text{otherwise,} \end{cases} \qquad (9)$$

and aggregate per site and voltage level while keeping evidence and estimate apart,

$$c_{\text{est}}(s,v) = \sum_{w \in L(s,v)} \hat{c}(w), \qquad c_{\text{sum}}(s,v) = \sum_{w:\,\text{evidence}} \hat{c}(w). \qquad (10)$$

Proposition 2 (Lower bound): Under (A1) a present `circuits` tag is correct and (A2) a three-phase circuit uses at least three conductors,

$$c_{\text{sum}}(s,v) \ \leq \ c_{\text{est}}(s,v) \ \leq \ c_{\text{true}}(s,v). \qquad (11)$$

Proof: Termwise, $\hat{c}(w) \leq c_{\text{true}}(w)$: with a tag this is (A1); with only `cables` the floor of $n_{\text{cables}}/3$ under-counts by (A2); and an existing way carries at least one circuit. Summation preserves the inequality, and c_{sum} omits non-negative terms of c_{est}. ■

The practical consequence is that a SubSLD figure states “at least this many circuits exist”. When such counts are used downstream—for corridor capacity proxies, for instance—the error is on the conservative side. Conductor bundles (`wires`) are treated identically.

VI. Three-Valued Direction Inference

Whether a line feeds a substation or is fed by it is not tagged. We order substations by their highest voltage level and infer

$$\hat{d}: \ G \longrightarrow \{\,\text{in},\ \text{out},\ \perp\,\} \qquad (12)$$

for each line group g with remote set far(g):

$$\hat{d}(g) = \text{in} \iff \exists s' \in \text{far}(g): \ kv_{\max}(s') > kv_v \quad \vee \quad kv_v = kv_{\text{top}}(s), \qquad (13)$$

$$\hat{d}(g) = \perp \iff \text{far}(g) = \varnothing, \qquad (14)$$

$$\hat{d}(g) = \text{out} \quad \text{otherwise.} \qquad (15)$$

far(g) is resolved from derived site-to-site connections, falling back to line names of the form “A–B line” when connections are missing. The third value is the point: when the remote end is unknown—overwhelmingly because the remote substation itself is absent from OSM—the method abstains and the figure renders the stub in grey. The abstention rate is reported as an evaluation metric, not suppressed.

TABLE II
NATIONWIDE EXTRACTION (ALL OF JAPAN, 10 REGIONS)

Quantity	Value
Substation structures extracted	7,239
Evidence-bound terminals	47,979
Busbar sections (observed)	2,559
Busbar sections (inferred, marked)	2,669
Bays	8,753
Transformers (structural)	2,590
Site-to-site connections derived	11,586
Extraction time (whole country)	≈ 4 s
Paired-figure rendering	1–6 s per site

VII. IMPLEMENTATION AND RESULTS

The pipeline has three stages: extraction (Algorithm 1), property aggregation (Section V), and rendering. All stages are deterministic and idempotent and are embedded in the project's single regeneration command, so the entire layer is reproducible in one pass.

Table II summarises the national instantiation. Rendering the whole country produced paired figures for every site; on a 160-thread server with ten regional workers the full sweep completed in about one hour with zero failures.

Fig. 2 shows a representative output. The left pane is the audit surface—one can see which line the diagram is talking about and how strongly it is attached—while the right pane is the electrical reading of the same evidence.

A. Coverage

Fig. 3 reports what the method could and could not observe. Circuit evidence (a `circuits` or `cables` tag) exists for 68.2 % of 40,087 lines; conductor-bundle tags for only 12.7 %. Terminal binding is dominated by the weakest class: 55.0 % lead-in, 29.3 % polygon, 15.7 % vertex. Direction inference returns in for 55.4 % and out for 5.2 % of 18,851 line groups and abstains on 39.4 %.

The sharpest limitation is busbar mapping: only 14.2 % of sites have a busbar way, with a strong regional gradient—53.3 % in Hokkaido against 5.1 % in Okinawa and 5.2 % in Tokyo (Fig. 3(c)). Point-geometry sites, which cannot host in-yard ways at all, are mapped at 3.2 %.

B. Photointerpretation of the busbar gap

Is the gap a mapping omission or a physical absence? We reviewed fourteen transmission substations that lack busbar ways, using their own SubSLD GeoPanes over aerial imagery. Nine (64 %) show outdoor air-insulated busbars and bay rows plainly visible from above—mappable omissions. One has in-yard ways and vertex-shared terminals but no `line=busbar` tag, i.e. a tagging omission repairable without new geometry. The remaining five are gas-insulated or indoor installations in which no busbar exists as a visible line and for which the inferred-busbar mechanism is the appropriate answer.

This diagnosis converts a coverage statistic into work: the nine mappable sites, together with a transmission line found by satellite review but absent from OSM, were published as a contribution list with per-site edit links.

TABLE III
MAPPING OF THE EXTRACTED STRUCTURE ONTO CIM/CGMES, WITH NATIONAL COUNTS

SubSLD element	CIM/CGMES class	Exported
SubstationSite	`cim:Substation`	7,240
VoltageLevel	`cim:VoltageLevel`	per site, per class
BusbarSection	`cim:BusbarSection`	4,743
of which inferred	(flagged, see text)	2,289
Bay	`cim:Bay`	8,475
TransformerSpec	`cim:PowerTransformer`	2,590
Terminal	`cim:Terminal`	with `ConnectivityNode`

VIII. STANDARDS-CONFORMANT EXPORT

A method that produces structure only for its own renderer has limited reach. The data model of Section IV was therefore defined against CIM from the outset, and each class maps one-to-one onto its CGMES counterpart (Table III). Exporting is consequently a direct serialisation rather than a translation.

Two points of discipline carry over into the export. First, a bay touching two busbar sections of the same voltage level is a coupler candidate, and this is disclosed in the object name rather than asserted as a breaker, because no switch was observed. Second—and more important for a consumer of the file—*inferred busbars remain marked*. CGMES 2.4 has no attribute for "estimated", so the provenance is written into `IdentifiedObject.description`:

```
inferred-topology (SubSLD): no busbar way in
OSM;
existence derived from strongly-bound
terminals
```

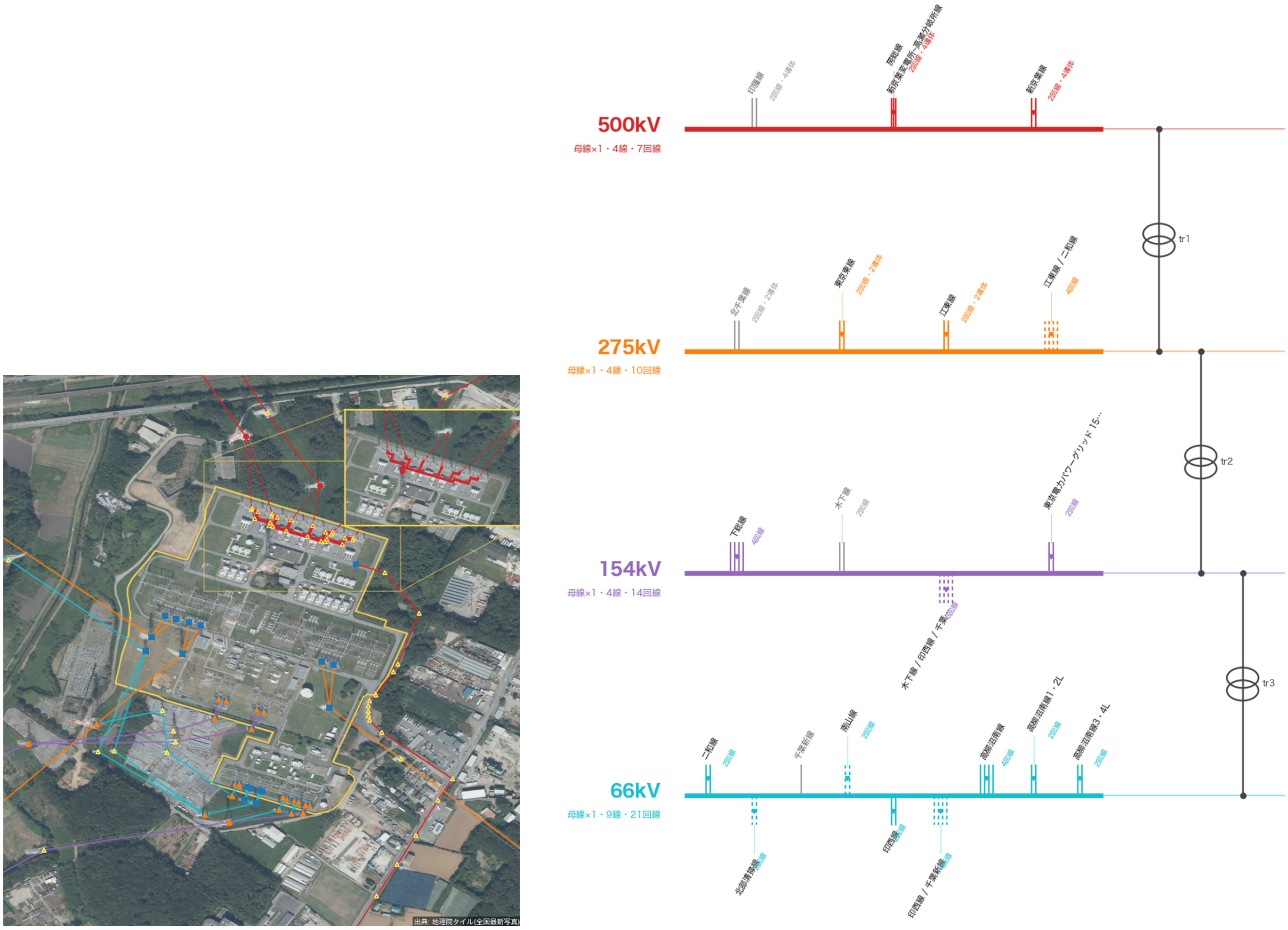
A tool that ignores the field still receives a valid, solvable topology; a tool or an analyst that reads it can separate the 2,454 observed busbars from the 2,289 inferred ones. The same principle governs transformer ratings elsewhere in the project: `ratedS` is written only where a nameplate source exists, never synthesised to fill the attribute.

The export runs as one stage of the regeneration pipeline over all ten regions, producing CGMES EQ and GL profiles; a Level-2 export (EQ/TP/SSH/SV) exists in the same project and has been round-tripped through an independent CIM importer to confirm the resulting cases still solve. The node-breaker layer introduced here rides on that established path.

IX. DISCUSSION

Showing the gap is part of the result. A model that silently completes missing structure produces a figure that looks finished and is therefore unauditable. SubSLD instead propagates evidence strength into the rendering: dashed stubs for lead-in bindings, grey for abstained direction, dashed busbars for inference. A reader can see, per element, how much to trust the drawing—and the same visual grammar identifies where the open database should be improved.

Limits. The lead-in band admits nearby passing lines as weak evidence; this is visible but not eliminated. Direction inference is a voltage-hierarchy heuristic, not a power-flow result, and is labelled as inference throughout. Conductor-count coverage is low enough that bundle information should



(a) GeoPane

(b) SLDPane

Fig. 2. Evidence-paired figure for a 500/275/154/66 kV substation (Shin-Keiyo). (a) Site outline, in-yard busbar geometry, real line geometry by voltage class, and binding markers over aerial imagery. (b) Machine-drawn single-line diagram: busbar sections, one stroke per circuit, dashed stubs for lead-in evidence, grey for abstained direction, transformers between levels. Imagery: GSI seamless photo tiles.

be treated as sparse annotation rather than a field. Finally, all coverage figures are snapshots of a living database.

Outlook. Three directions follow naturally. Automatic detection of towers and busbars from imagery would raise the evidence base without human review. The contribution list closes a loop in which the model improves its own source. And a bay-level structure with inferred busbars is the precondition for breaker-level analyses on an open national model.

X. Conclusion

We have shown that the internal composition of substations can be reconstructed nationwide from volunteered geographic information, provided extraction is formulated so that soundness holds by construction, estimation is certified as a lower bound, and inference is allowed to abstain. Applied to Japan, SubSLD produced structures for 7,239 substations with 47,979 evidence-bound terminals and an evidence-paired figure for every site, alongside an honest account of what open data does not yet contain. The measured gaps are not a postscript to the result—they are the part of it that tells the community what to map next.

Data and Code Availability

The method, generators, viewer and datasets are part of the All-Japan-Grid project (https://github.com/lutelute/All-Japan-Grid), released as `v1.8.0`; CGMES EQ/GL exports including the node-breaker layer are regenerated by `scripts/export_cim.py`. The nationwide viewer is at https://lutelute.github.io/All-Japan-Grid/subsld.html. Aerial imagery is from the Geospatial Information Authority of Japan; base map data © OpenStreetMap contributors (ODbL).

Acknowledgment

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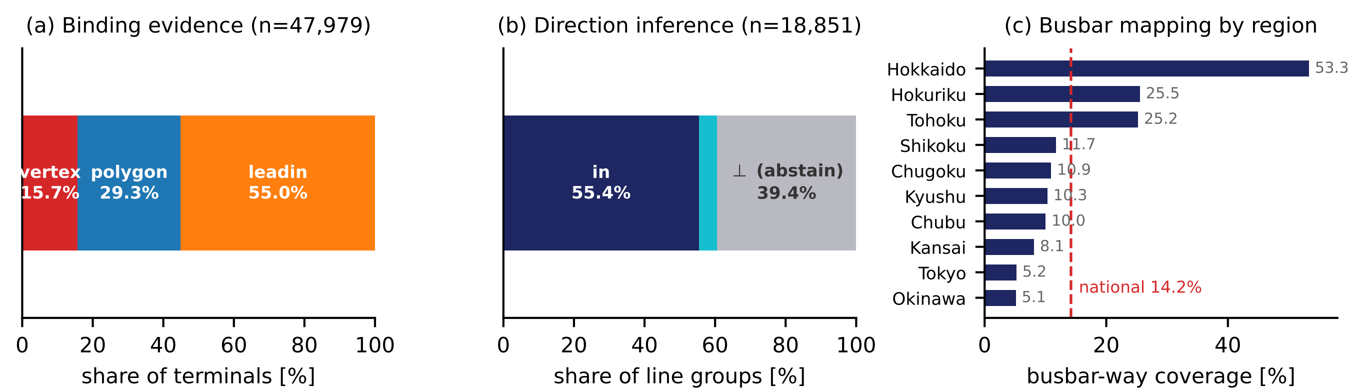

Fig. 3. Coverage measured on the national extraction. (a) Distribution of terminal binding evidence. (b) Outcome of direction inference, including explicit abstention. (c) Busbar-way mapping rate by region, against the national mean.